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① Title: Ginger-derived extracellular vesicles attenuate aspiration-induced lung injury in mice: involvement of miR-339-3p

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Abstract:	② Abstract Background Aspiration-induced acute lung injury (ALI) is a serious perioperative complication characterized by excessive inflammation and oxidative stress. Ginger-derived extracellular vesicles (GiEVs) possess anti-inflammatory and antioxidant properties. This study investigated the therapeutic effects of GiEVs and explored the contribution of their microRNA cargo in aspiration-induced ALI. Methods Adult male C57BL/6 mice were subjected to oropharyngeal instillation of a gastric-content mimic to induce ALI and received intratracheal GiEV treatment (ALI+GiEV) or vehicle (ALI). Sham and GiEV-only groups served as controls. After 48 hours, lung injury, pulmonary function, and inflammatory, oxidative, and apoptotic responses were assessed. Results

	Compared with the Sham group, ALI mice developed significant lung injury, characterized by histopathological damage, impaired pulmonary function, pulmonary edema, leukocyte infiltration, and increased lung injury scores. These changes were accompanied by increased inflammatory signaling (phosphorylated nuclear factor-κB, inducible nitric oxide synthase, and pro-inflammatory cytokines), elevated oxidative stress, and enhanced apoptosis. GiEV treatment significantly improved pulmonary function and attenuated lung injury, inflammation, oxidative stress, and apoptosis compared with the ALI group. Small RNA sequencing identified microRNA-339-3p (miR-339-3p) as the most abundant GiEV cargo. Inhibition of miR-339-3p attenuated the protective effects of GiEVs on lung injury and pulmonary edema. Conclusion GiEVs attenuate aspiration-induced ALI through coordinated suppression of inflammation, oxidative stress, and apoptosis. miR-339-3p contributes, at least in part, to these protective effects.
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Dear Professor Sangseok Lee,

We are pleased to submit our manuscript entitled

"Ginger-derived extracellular vesicles attenuate aspiration-induced lung injury in mice: involvement of miR-339-3p"

for consideration as an *Experimental Research* article in the ***Korean Journal of Anesthesiology***. Aspiration-induced acute lung injury (ALI) remains a major cause of respiratory failure in perioperative and critical care settings, with no effective targeted therapies currently available. In this murine study, we demonstrate that **intratracheal administration of ginger-derived extracellular vesicles (GiEVs)** significantly attenuates lung injury and inflammatory responses. We further identify **miR-339-3p as a key mediator** of these protective effects, as its inhibition abolishes the therapeutic benefit.

This work builds upon our prior findings that **human placental MSC-derived extracellular vesicles (hPMSC-EVs)** mitigate aspiration-induced ALI. However, the clinical translation of MSC-derived EVs is limited by scalability, variability, and regulatory challenges. To address these limitations, we investigated **plant-derived EVs** as a more scalable and potentially safer alternative. Our findings demonstrate comparable therapeutic efficacy and provide mechanistic insight into a microRNA-mediated pathway, supporting the development of **next-generation, cell-free EV-based therapies**.

This manuscript has not been published or submitted elsewhere. ChatGPT was used solely to improve language clarity, and all content has been reviewed and approved by the authors. The authors declare no conflicts of interest.

We believe this study will be of interest to readers in anesthesiology, critical care, and translational lung injury research.

Thank you for your consideration and assistance.

Respectfully submitted,

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Title page

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③ **Running title**: GiEVs vs. aspiration lung injury

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⑥ **Conflict of interest:** The authors declare that neither they nor any immediate family members have any financial or personal interest in any organization or entity with a direct stake in the subject matter or materials discussed in this article.

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The animal studies of the present study received ethical approval from the Institutional Animal Care and Use Committee of Taipei Medical University, Taipei, Taiwan (Title: The therapeutic effects of preconditioned placenta-derived mesenchymal stem cell-derived exosomes in aspiration pneumonia-related lung injury and its possible mechanism. Approval number: LAC-2022-0427); approval date: April 06, 2023).

⑩ **Research registration number**: Not applicable.

Data availability: All data supporting the findings of this study are reported within this article and its supplementary information. The raw data of small RNA-seq analysis of ginger-derived extracellular vesicles have been deposited in Harvard Dataverse (https://doi.org/10.7910/DVN/HAKU9D), and are available from the corresponding author upon reasonable request.

① **Title**: Ginger-derived extracellular vesicles attenuate aspiration-induced lung injury in mice: involvement of miR-339-3p

Running Title: GiEVs vs. aspiration lung injury

② **Abstract:**

Background: Aspiration-induced acute lung injury (ALI) is a serious perioperative complication characterized by excessive inflammation and oxidative stress. Ginger-derived extracellular vesicles (GiEVs) possess anti-inflammatory and antioxidant properties. This study investigated the therapeutic effects of GiEVs and explored the contribution of their microRNA cargo in aspiration-induced ALI.

Methods: Adult male C57BL/6 mice were subjected to oropharyngeal instillation of a gastric-content mimic to induce ALI and received intratracheal GiEV treatment (ALI+GiEV) or vehicle (ALI). Sham and GiEV-only groups served as controls. After 48 hours, lung injury, pulmonary function, and inflammatory, oxidative, and apoptotic responses were assessed.

Results: Compared with the Sham group, ALI mice developed significant lung injury, characterized by histopathological damage, impaired pulmonary function, pulmonary edema, leukocyte infiltration, and increased lung injury scores. These changes were accompanied by increased inflammatory signaling (phosphorylated nuclear factor-κB, inducible nitric oxide synthase, and pro-inflammatory cytokines), elevated oxidative stress, and enhanced apoptosis. GiEV treatment significantly improved pulmonary function and attenuated lung injury, inflammation, oxidative stress, and apoptosis compared with the ALI group. Small RNA sequencing identified microRNA-339-3p (miR-339-3p) as the most abundant GiEV cargo. Inhibition of miR-339-3p attenuated the protective effects of GiEVs on lung injury and pulmonary edema.

Conclusion: GiEVs attenuate aspiration-induced ALI through coordinated suppression of inflammation, oxidative stress, and apoptosis. miR-339-3p contributes, at least in part, to these protective effects.

Keywords: aspiration; lung injury; ginger-derived extracellular vesicles; mice; miR-339-3p

③ Introduction

Aspiration-induced acute lung injury (ALI) remains a significant cause of respiratory failure in perioperative and intensive care settings, particularly among patients undergoing anesthesia, airway manipulation, or mechanical ventilation [1-3] The pathophysiology of aspiration-induced ALI is characterized not only by the direct chemical injury caused by aspirated gastric contents but also by an exaggerated host inflammatory response that amplifies pulmonary damage. This inflammatory cascade contributes to alveolar epithelial injury, impaired gas exchange, and respiratory failure [1-3]. Despite advances in antimicrobial therapy and supportive respiratory care, no targeted therapies currently exist to effectively modulate the inflammatory pathways driving aspiration-induced ALI [3-5].

Extracellular vesicles (EVs) have recently emerged as promising modulators of intercellular communication and inflammatory signaling in acute lung injury [6-8]. EVs derived from mesenchymal stem cells (MSCs) have demonstrated protective effects in several preclinical models of lung injury by delivering regulatory microRNAs and other bioactive molecules that suppress inflammatory pathways [6-8]. In our previous work, EVs derived from human placental MSCs were shown to attenuate aspiration-induced ALI [9]. However, the clinical translation of MSC-derived EV therapies remains limited, mainly due to the restricted availability and scalability of biological sources [10, 11].

To address this limitation, plant-derived EVs have emerged as a scalable and cost-effective alternative [12, 13]. These EVs can be readily isolated in large quantities from edible plants and are enriched with bioactive lipids and small RNAs capable of mediating cross-kingdom regulatory interactions [12, 13]. Among these, ginger-derived EVs (GiEVs) have been reported to exhibit potent anti-inflammatory and immunomodulatory properties [13]. Their abundant availability and favorable production scalability make GiEVs a promising alternative to mammalian cell-derived EVs for therapeutic development.

To further investigate the therapeutic potential of GiEVs in aspiration-induced ALI, we conducted this murine study to test the hypothesis that GiEV administration could mitigate aspiration-induced lung injury in adult male mice. In addition to evaluating their protective effects, we examined the impact of GiEVs on inflammatory, oxidative, and apoptotic responses in lung tissues and explored the potential contribution of their microRNA cargo.

④ Materials and methods

A. Isolation and physical characterization of GiEVs

GiEVs were purified through a standardized differential ultracentrifugation technique [14]. Briefly, fresh ginger rhizomes were homogenized in chilled phosphate-buffered saline (PBS) at 4°C using a high-speed blender, maintaining a temperature below 10°C to minimize enzymatic degradation. The resulting homogenate was passed through 100-μm mesh filters (Millipore, Burlington, MA, US) to exclude large fibrous debris. The filtrate was then clarified via consecutive centrifugation steps at 500 × g for 10 minutes and 2,000 × g for 20 minutes (LYNX 6000 Superspeed Centrifuge, Thermo Fisher Scientific, Waltham, MA, USA) to strip away cellular remnants. The supernatant was further processed at 10,000 × g for 30 minutes (LYNX 6000 Superspeed Centrifuge) to eliminate organelle contaminants. Final isolation was achieved by ultracentrifugation at 150,000 × g for 120 minutes at 4°C using an Optima L-90k system (Beckman Coulter, Brea, CA, USA). GiEV morphology was visualized using transmission electron microscopy (TEM; Hitachi HT-7700, Tokyo, Japan), while their particle size distribution and particle concentration were quantified through nanoparticle tracking analysis (NTA; NanoSight NS300, Malvern Panalytical, Malvern, UK) [14-16] and surface potential was measured by the zeta potential assay (Zetasizer Pro; Malvern Panalytical) [17].

B. In vivo pulmonary distribution analysis

To evaluate pulmonary delivery efficiency, GiEVs labeled with Cy7 mono NHS ester (Amersham, Buckinghamshire, UK) were administered intratracheally to mice at a dose of 1×10^9 particles per mouse, as adapted from our previous protocol [9]. Following anesthesia and euthanasia by decapitation, organs (heart, lungs, liver, kidneys, spleen, and bladder) were harvested at 0, 2, 24, and 48 hours post-instillation. Biodistribution was evaluated using an in vivo imaging system (IVIS Lumina XRMS; PerkinElmer, Waltham, MA, USA) to detect Cy7 fluorescence, and quantitative analysis of signal intensity was performed using Living Image software (PerkinElmer), according to the manufacturer's manual. Based on the observed retention in the lung tissue, a 24-hour dosing interval was selected for subsequent therapeutic experiments.

C. Animals and ethical statement

Adult male C57BL/6 mice (7 weeks old) were obtained from the National Laboratory Animal Center (Taipei, Taiwan). Only male mice were used in this study because clinical observations suggest higher mortality from aspiration pneumonia in males than in females (odds ratio 2.21) [18]. Animals were housed under standard laboratory conditions (12-hour light/dark cycle) with ad libitum access to water and standard chow. All experimental procedures were approved by the Institutional Animal Care and Use Committee of Taipei Medical University (IACUC approval number: LAC-2022-0427) and complied with the Guide for the Care and Use of Laboratory Animals (National Institutes of Health, USA).

D. Murine aspiration-induced ALI model

A murine model of aspiration-induced ALI was established via oropharyngeal installation of a gastric-content mimic, as previously described by our group [9]. The gastric content–mimic solution consisted of lipopolysaccharide (2.5 mg/mL; L3129, Sigma-Aldrich, St. Louis, MO, USA), pepsin (2 mg/mL; P6887, Sigma-Aldrich, St. Louis, MO, USA), and xanthan gum thickener (12 mg/mL; ThickenUp®, Nestlé Health Science, Lutry, Switzerland). The mixture was adjusted to pH 1.6 with hydrochloric acid (HCl; Sigma-Aldrich) before administration. Mice were anesthetized with 3% isoflurane (Terrell, Piramal Critical Care, Bethlehem, PA, USA) and placed in a supine position with the tongue gently extended to facilitate the delivery of the gastric content–mimic solution into the posterior oropharynx. This procedure was designed to simulate macroaspiration of acidic gastric contents into the lower airways

E. Experimental grouping

In the independent cohort, seventy-two mice were randomly allocated to four experimental groups (n = 18 per group): Sham, GiEV, ALI, and ALI-GiEV. The ALI and Sham groups received oropharyngeal installation of the gastric-content mimic and phosphate-buffered saline (PBS; Sigma-Aldrich), respectively. These groups subsequently received two intratracheal administrations of PBS at 2 and 26 hours after aspiration. The ALI-GiEV and GiEV groups underwent the same initial treatments (gastric-content mimic or PBS, respectively), followed by two intratracheal administrations of GiEVs (1×10^9 particles per mouse) at 2 and 26 hours post-aspiration. Animals were monitored for 48 hours after aspiration or for an equivalent duration in the Sham and GiEV groups. Surviving mice were then anesthetized and euthanized by decapitation, and lung injury severity and mechanistic endpoints were subsequently assessed. All experimental procedures and outcome assessments were conducted by investigators blinded to group allocation.

To determine the therapeutic dose of GiEVs, a preliminary dose-escalation study was conducted in an independent cohort. GiEVs were administered intratracheally at doses of 1×10^8, 1×10^9, or 1×10^{10} particles per mouse in aspiration-induced ALI mice (n = 4 per dose). Additional mice receiving sham operation or aspiration-induced ALI (n = 4 per group) served as controls. Leukocyte infiltration was assessed by total white blood cell (WBC) counts in bronchoalveolar lavage fluid at

48 h. Doses of 1×10^9 and 1×10^{10} particles significantly reduced leukocyte infiltration, whereas 1×10^8 particles had no effect (Supplementary Fig. 1A). Based on these results, 1×10^9 particles per mouse was selected for subsequent experiments and for the in vivo pulmonary distribution analysis described in Section 2.2.

F. Assessment of lung injury and function alterations

F.1. Histopathological evaluation and lung injury scoring

The first cohort of six mice per group were used for this assay. Briefly, a tracheostomy was performed, and a tracheostomy tube (22-gauge intravenous catheter; BD Insyte, BD Biosciences, Franklin Lakes, NJ, USA) was inserted into the trachea to facilitate the procedure. The left bronchus was ligated, and the right lungs were infused with 4% formaldehyde (Servicebio, Wuhan, China) at constant pressure via the tracheostomy tube before excision and paraffin embedding (Sigma-Aldrich). Lung sections were stained with hematoxylin and eosin (H&E; 245880, Abcam, Cambridge, UK) and digitized at 200× magnification using a high-resolution slide scanner (MoticEasyScan Pro 6; Motic Asia). Lung injury was evaluated using a standardized semi-quantitative scoring system by a pathologist blinded to group allocation. The scoring system assessed five parameters: (1) neutrophils in the alveolar space, (2) neutrophils in the interstitial space, (3) hyaline membrane formation, (4) proteinaceous debris in the airspaces, and (5) alveolar septal thickening [9, 16]. Each parameter was graded on a scale of 0–2 (0 = normal, 2 = severe injury). Six random microscopic fields per section were analyzed, and the scores were averaged to generate the final lung injury score for each section.

F.2. Quantification of pulmonary edema

Following right lung harvesting as described above, the upper lobe of the left lung from the same cohort of six mice per group was collected and immediately weighed to obtain the wet weight. The tissue was then dried in an oven at 80 °C for 24 h until a stable dry weight was achieved. The wet-to-dry weight ratio was calculated as an index of pulmonary edema [9, 16]. The lower lobe of the left lung was snap-frozen in liquid nitrogen and stored at −80 °C for subsequent molecular and biochemical analyses.

F.3. Respiratory function analysis

A second cohort of six mice per group was used for respiratory function analysis. Anesthesia, tracheostomy, and tracheostomy tube insertion were performed as described above. The tracheostomy tube was then connected to a small-animal ventilator (SAR-1000; CWE Inc., Ardmore, PA, USA) integrated with the flexiWare 8 system (SCIREQ Inc., Montreal, QC, Canada). Pulmonary function parameters—including inspiratory volume, expiratory volume, dynamic compliance, airway resistance, forced expiratory volume, and elastance—were measured at a respiratory rate of 150 breaths per minute and a tidal volume of 0.2 mL [9, 16].

F.4. Leukocyte infiltration analysis

A third cohort of six mice per group was used for this assay. Briefly, after anesthesia, tracheostomy and insertion of a tracheostomy tube, the lungs were lavaged five times with 1 mL aliquots of sterile PBS to collect bronchoalveolar lavage fluid (BALF) [9, 16]. The BALF samples were fixed with 4% formaldehyde (Sigma-Aldrich, St. Louis, MO, USA). Total WBC counts and differential leukocyte profiles (neutrophils, lymphocytes, and monocytes) were determined using a ProCyte Dx automated hematology analyzer (IDEXX Laboratories, Westbrook, ME, USA). These measurements were used to assess leukocyte infiltration into the alveolar space [9, 16].

G. Inflammatory molecular analyses

Molecular signaling associated with pulmonary inflammation, including nuclear factor-κB (NF-κB, the crucial upstream transcription factor) [19] and inducible nitric oxide synthase (iNOS, the macrophage M1 phase polarization marker) [20], was examined through immunoblotting blotting assay and immunohistochemistry assay. In addition, expression of pro-inflammatory cytokines was measured using enzyme-linked immunosorbent assay (ELISA) [9, 16].

G.1. Immunoblotting assay

Immunoblotting assay was conducted following published protocols [9, 16]. In brief, lung homogenates from snap-frozen tissues (4 mice per group) were processed for total protein extraction using RIPA lysis buffer containing protease and phosphatase inhibitors (Thermo Fisher Scientific, Waltham, MA, USA). Protein concentrations were determined using a bicinchoninic acid (BCA) protein assay kit (Thermo Fisher Scientific). Equal amounts of protein (20 μg per lane) were separated by SDS–polyacrylamide gel electrophoresis and transferred onto polyvinylidene difluoride (PVDF) membranes (IPVH00010; Millipore). The membranes were blocked with 5% non-fat milk in Tris-buffered saline containing 0.1% Tween-20 (TBST; Sigma-Aldrich) for 1 hour at room temperature and then incubated overnight at 4 °C with primary antibodies against phosphorylated NF-κB p65 (p-NF-κB p65 Ser536; diluted at 1:5,000; Cell Signaling Technology, Danvers, MA, USA), inducible nitric oxide synthase (iNOS; ab15323, 1:5,000; Abcam, Cambridge, UK), and Actin (1:5,000; Sigma-Aldrich) as the loading control. After washing with TBST, membranes were incubated with horseradish peroxidase–conjugated secondary antibodies (7074; Cell Signaling Technology, Danvers, MA, USA) for 1 hour at room temperature. Protein bands were visualized using an enhanced chemiluminescence (ECL) detection system (WBKLS0500, Millipore) and imaged using a chemiluminescence imaging system (Bio-Rad Laboratories, Hercules, CA, USA). Band intensities were quantified using ImageJ software (NIH, USA), and the relative expression of target proteins was normalized to Actin.

G.2. Immunohistochemistry assay

Immunohistochemistry assay was conducted also following published protocols [9,16]. For immunohistochemistry, paraffin-embedded lung sections from six mice per group were incubated with primary antibodies against phosphorylated NF-κB p65 (p-NF-κB p65 Ser536; 1:200; Cell

Signaling Technology) or inducible nitric oxide synthase (iNOS; ab15323, 1:200; Abcam). Images were analyzed using ImageJ software to determine the mean staining intensity of NF-κB and iNOS from six randomly selected fields per section.

G.3. ELISA

As above-mentioned, levels of pro-inflammatory cytokines in snap-frozen lung tissues (six mice per group) were measured using ELISA. Lung tissue samples were prepared according to previously established procedures [9,16]. The concentrations of tumor necrosis factor-α (TNF-α), interleukin-1β (IL-1β), and interleukin-6 (IL-6) were determined using commercially available ELISA kits (Enzo Life Sciences, Farmingdale, NY, USA).

H. Oxidative status assay

Oxidative stress in lung tissues was evaluated by assessing malondialdehyde (MDA), a marker of lipid peroxidation [21], and heme oxygenase-1 (HO-1), an enzyme induced under oxidative stress conditions [22], using immunoblotting and immunohistochemical analyses. Following the procedures described above, protein expression of MDA and HO-1 was analyzed by immunoblotting using snap-frozen lung tissues from four mice per group. PVDF membranes were incubated with primary antibodies against MDA (ab27642, 1:5,000; Abcam), HO-1 (ab13248, 1:5,000; Abcam), or Actin (A2066, 1:5,000; Sigma-Aldrich) as the loading control.
For immunohistochemical analysis, paraffin-embedded lung sections from six mice per group were incubated with primary antibodies against MDA (ab27642, 1:200; Abcam) or HO-1 (ab13248, 1:200; Abcam), following the procedures described above.

I. Apoptosis assay

Apoptosis in lung tissues was evaluated by detecting DNA fragmentation in paraffin-embedded lung sections from six mice per group using a terminal deoxynucleotidyl transferase dUTP nick-end labeling (TUNEL) assay (Roche Diagnostics, Mannheim, Germany), according to the manufacturer's protocols. The proportion of apoptotic cells was calculated as the ratio of TUNEL-positive nuclei to total DAPI-stained nuclei (Sigma-Aldrich), using confocal laser scanning microscopy (LSM-780; Zeiss, Oberkochen, Germany). Activation of the apoptotic pathway was further assessed by immunoblotting analysis of cleaved caspase-3 [24] in snap-frozen lung tissues from four mice per group. PVDF membranes were incubated with primary antibodies against cleaved caspase-3 (ab2302, 1:500; Abcam) and Actin (A2066, 1:5,000; Sigma-Aldrich), following the procedures described above.

J. Small RNA sequencing and bioinformatics of GiEV cargos

Total RNA was isolated from GiEVs using the miRNeasy Micro Kit (QIAGEN, Hilden, Germany), following published protocols [15]. For small RNA library preparation, 100 ng of total RNA was ligated to 3′ and 5′ adapters using the QIAseq® miRNA Library Kit (QIAGEN). Adapter-ligated RNA fragments were subsequently reverse-transcribed with the QIAseq miRNA NGS RT Enzyme and amplified by polymerase chain reaction (PCR). The amplified libraries were size-selected (170–200 bp) using QIAseq beads to enrich for miRNA-containing fragments. The purified libraries were sequenced on an Illumina NovaSeq 6000 platform (Illumina, San Diego, CA, USA) to generate 75-bp single-end reads and sequencing was performed (Genomics, BioSci & Tech Co., Ltd., New Taipei City, Taiwan). Raw sequencing reads were processed by adaptor trimming using TrimGalore! (v0.6.6; Babraham Bioinformatics, Cambridge, UK) and aligned to reference microRNA sequences from miRBase (v22.1) using Bowtie (v1.3.0; Johns Hopkins University, Baltimore, MD, USA). Read counts were normalized using edgeR (v3.26.5; Bioconductor, Cambridge, MA, USA), and differentially expressed microRNAs were identified with DEGSeq (v1.48.0; Bioconductor, Cambridge, MA, USA).

K. Preparation of microRNA inhibitor–loaded GiEVs

To functionally inhibit the key microRNA identified in Section 2.10, a specific microRNA inhibitor was designed and synthesized. The inhibitor was introduced into GiEVs by electroporation (150 V, 100 μF) using a Gene Pulser Xcell electroporation system (Bio-Rad Laboratories, Hercules, CA, USA) [9, 15, 16]. Following electroporation, the samples were incubated to allow membrane recovery. Unincorporated inhibitors were subsequently removed, and the loaded GiEVs were re-isolated by ultracentrifugation. The resulting inhibitor-loaded GiEV pellet was resuspended in sterile PBS and stored at −80 °C until further use.

L. Functional validation of the identified microRNA

An independent cohort of 36 mice was randomly assigned to three experimental groups (n = 12 per group): ALI, ALI–GiEV, and ALI–GiEVi. All animals underwent oropharyngeal instillation of the gastric content–mimic solution to induce aspiration-induced-ALI. At 2 and 26 h after aspiration, mice received intratracheal administration of PBS, GiEVs (1×10^9 particles), or microRNA inhibitor–loaded GiEVs (GiEVi; 1×10^9 particles), respectively. Animals were monitored for 48 hours, after which lung injury severity was assessed by histopathological analysis and pulmonary edema was evaluated in surviving mice.

M. Blinding experimental designs

Blinding measures were implemented to minimize bias. Personnel responsible for inducing the aspiration model and administering GiEVs or PBS were not involved in outcome assessment. Sample collection, including lung harvesting and BALF retrieval, and all outcome assessments (histopathology, wet-to-dry weight ratio, pulmonary function testing, ELISA, immunoblotting, and immunohistochemistry) were performed by investigators blinded to group allocation. Samples were assigned randomized identification codes, and treatment allocation was not disclosed until all data collection and analyses were completed.

N. Sample size and statistical analysis

Sample size estimation was conducted using G*Power version 3.1.9.7 (Heinrich Heine University Düsseldorf, Germany) based on preliminary data from BALF total WBC counts. In the pilot study, the ALI group exhibited a mean WBC count of 6.05 ± 1.61 K/μL, whereas the GiEV-treated group showed a mean of 2.55 ± 0.81 K/μL, corresponding to a Cohen's d of 2.75. To avoid overestimation of effect size and account for biological variability, a conservative effect size of 1.2 was used for the final sample size calculation. Power analysis under a one-way ANOVA framework with four groups ($\alpha = 0.05$, power = 0.80) indicated that six mice per group were sufficient to achieve adequate statistical power. In accordance with the 3Rs principles (Replacement, Reduction, and Refinement) of ethical animal research [25], the minimum number of animals required was used. Accordingly, six mice per group were included for the primary in vivo analyses unless otherwise specified. For molecular analyses, a subset of samples was used based on tissue availability and assay requirements. Immunoblotting was performed using four mice per group, while ELISA and TUNEL assays included six mice per group. All experiments were independently repeated at least three times. Data are presented as mean ± standard deviation. Between-group comparisons were performed using one-way ANOVA followed by Tukey's post hoc test. A two-tailed P value < 0.05 was considered statistically significant. Statistical analyses were conducted using GraphPad Prism software (GraphPad Software, San Diego, CA, USA).

⑤ Results

Characterization and biodistribution of GiEVs

Morphological assay by TEM revealed that the isolated GiEVs exhibited a characteristic round or cup-shaped morphology with a bilayer membrane structure, consistent with plant-derived EVs (Figure 1A). Sizing assay using NTA showed a size distribution peaking at 222.3 nm (Figure 1B). The zeta potential of GiEVs was −10.22 mV, indicating favorable colloidal stability (Figure 1C).

In vivo biodistribution analysis using IVIS demonstrated that GiEVs were predominantly localized in the lungs following intratracheal administration (Figure 1D). Quantitative analysis showed that lung fluorescence intensity significantly increased at 2 hours compared with baseline (0 hours) ($p = 0.026$), followed by a gradual decline. Signal intensity at 24 hours was not significantly different from baseline ($p = 0.989$), and fluorescence at 48 hours was minimal and comparable to baseline ($p = 0.999$). Compared with the peak at 2 hours, fluorescence intensity was significantly reduced at both 24 hours ($p = 0.039$) and 48 hours ($p = 0.028$), whereas no significant difference was observed between 24 and 48 hours ($p = 0.995$). In addition, fluorescence signals in other organs, including the heart, liver, kidneys, spleen, and bladder, were weak and transient, with low-level signals detected at 2 hours post-administration that were not significantly different from baseline and absent at later time points.

Notably, based on the lung biodistribution profile, a 24-hour dosing interval was selected for subsequent therapeutic experiments.

GiEVs attenuate lung injury and mitigate pulmonary functional impairment

Histopathological examination revealed marked lung injury in the ALI group following aspiration, characterized by alveolar wall thickening, inflammatory cell infiltration, and disruption of alveolar architecture. These pathological changes were markedly attenuated in mice treated with GiEVs (ALI+GiEV group), whereas normal lung histology was observed in the Sham and GiEV groups (Figure 2A). Consistently, lung injury scores were significantly higher in the ALI group than in the Sham group ($p < 0.001$), while scores were significantly reduced in the ALI+GiEV group compared with the ALI group ($p < 0.001$). Lung injury scores remained low in the Sham and GiEV groups (Figure 2A).

Pulmonary function testing also demonstrated substantial functional impairment in the ALI group. Compared with the Sham group, ALI mice showed significantly lower inspiratory volume, forced expiratory volume in 0.1 seconds ($FEV_{0.1}$), forced expiratory volume, and dynamic compliance (all $p < 0.001$), along with higher airway resistance and elastance (both $p < 0.001$) (Figure 2B). Treatment with GiEVs significantly improved inspiratory volume, $FEV_{0.1}$, forced expiratory volume, and dynamic compliance, while reducing airway resistance and elastance compared with the ALI group (all $p < 0.001$). In contrast, mice in the Sham and GiEV groups exhibited normal pulmonary function (Figure 2B).

GiEVs suppress pulmonary edema and leukocyte infiltration

Compared with the Sham group, mice in the ALI group exhibited a significantly higher lung wet-to-dry weight ratio ($p < 0.001$), indicating increased pulmonary edema (Figure 3A). This was accompanied by a marked increase in total WBC counts in BALF, along with significantly elevated neutrophil, lymphocyte, and monocyte counts (all $p < 0.001$) (Figure 3B). GiEV treatment significantly attenuated these changes, as evidenced by a reduced wet-to-dry weight ratio and decreased BALF WBC counts and differential cell counts in the ALI+GiEV group compared with the ALI group ($p = 0.003$, < 0.001, $= 0.001$, < 0.001, and < 0.001, respectively). In contrast, mice in the Sham and GiEV groups exhibited minimal pulmonary edema and leukocyte infiltration (Figure 3A & 3B).

GiEVs inhibit NF-κB signaling, macrophage activation and cytokine upregulation

Immunohistochemistry and immunoblotting analyses demonstrated increased expression of phosphorylated NF-κB and iNOS in the ALI group compared with the Sham group (Figure 4A &

4B). Quantitative analyses confirmed that p-NF-κB and iNOS levels were significantly elevated in the ALI group (immunohistochemistry: both $p < 0.001$; immunoblotting : both $p < 0.001$). In addition, pro-inflammatory cytokines, including TNF-α, IL-1β, and IL-6, were markedly increased in the ALI group compared with the Sham group ($p < 0.001$, $= 0.022$, and $= 0.001$, respectively) (Figure 4C). GiEV treatment attenuated these inflammatory responses, as evidenced by reduced p-NF-κB and iNOS expression on immunohistochemistry and immunoblotting analyses (Figure 4A & 4B). Quantitative analyses further demonstrated significantly lower levels of p-NF-κB, iNOS, TNF-α, IL-1β, and IL-6 in the ALI+GiEV group compared with the ALI group ($p < 0.001$, < 0.001, $= 0.001$, $= 0.004$, and $= 0.007$, respectively) (Figure 4A, 4B, & 4C). In contrast, mice in the Sham and GiEV groups exhibited minimal inflammatory activation.

GiEVs suppress pulmonary oxidative stress

Immunohistochemistry and immunoblotting analyses demonstrated increased MDA levels and HO-1 expression in the ALI group compared with the Sham group (Figure 5A & 5B). Quantitative analyses confirmed that MDA and HO-1 levels were significantly elevated in the ALI group (immunohistochemistry: both $p < 0.001$; immunoblotting: both $p < 0.001$, respectively). GiEV treatment mitigated oxidative stress, as evidenced by reduced MDA levels HO-1 expression on immunohistochemistry and immunoblotting (Figure 5A & 5B). Quantitative analyses further demonstrated significantly lower MDA levels and HO-1 expression in the ALI+GiEV group compared with the ALI group (immunohistochemistry: both $p < 0.001$; immunoblotting: both $p < 0.001$, respectively). In contrast, mice in the Sham and GiEV groups exhibited minimal MDA levels and HO-1 expression.

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GiEVs suppress pulmonary apoptosis

TUNEL staining revealed a marked increase in TUNEL-positive signals lung tissues in the ALI group compared with the Sham group (Figure 6A). Quantitative analysis confirmed that the number of TUNEL-positive cells was significantly higher in the ALI group compared with the Sham group ($p < 0.001$). Consistently, immunoblotting demonstrated increased expression of cleaved caspase-3 in the ALI group, which was further confirmed by quantitative analysis ($p < 0.001$) (Figure 6B). GiEV treatment attenuated apoptosis, as evidenced by a reduction in TUNEL-positive cells and decreased cleaved caspase-3 expression in the ALI+GiEV group compared with the ALI group (Figure 6A & 6B). Quantitative analyses further demonstrated significantly lower numbers of TUNEL-positive cells and reduced cleaved caspase-3 levels in the ALI+GiEV group (both $p < 0.001$). In contrast, minimal apoptotic activity was observed in the Sham and GiEV groups.

MicroRNA-339-3p mediates the therapeutic effects of GiEVs

Small RNA sequencing identified microRNA-339-3p (miR-339-3p) as the most abundant microRNA in GiEV cargo (Figure 7A). To assess its functional role, a loss-of-function approach was performed using GiEVs treated with a specific miR-339-3p inhibitor (5′-CGGCUCUGUCGAGGCGCUCA-3′; BioTool, New Taipei City, Taiwan), hereafter referred to as GiEVi.

Histopathological examination revealed marked alveolar wall thickening, inflammatory cell infiltration, and disruption of alveolar architecture in the ALI group (Figure 7B). These pathological changes were attenuated in mice treated with unmodified GiEVs (ALI+GiEV group), whereas inhibition of miR-339-3p largely abolished these protective effects, as evidenced by aggravated lung injury in the ALI+GiEVi group (Figure 7B). Consistently, lung injury scores were significantly reduced in the ALI+GiEV group compared with the ALI group ($p = 0.001$), while scores in the ALI+GiEVi group were significantly increased compared with the ALI+GiEV group ($p = 0.002$) (Figure 7B).

Similarly, GiEV treatment significantly reduced pulmonary edema, as indicated by a lower lung wet-to-dry weight ratio in the ALI+GiEV group compared with the ALI group (p = 0.005) (Figure 7C). In contrast, inhibition of miR-339-3p reversed this effect, resulting in a significantly higher wet-to-dry weight ratio in the ALI+GiEVi group compared with the ALI+GiEV group (p = 0.009) (Figure 7C).

These findings indicate that miR-339-3p is a key mediator of the protective effects of GiEVs in aspiration-induced acute lung injury.

⑥ Discussion

The present study confirms our hypothesis that GiEVs mitigate aspiration-induced lung injury in adult male mice. This protective effect was evidenced by consistent improvements across structural, functional, and molecular endpoints. GiEV treatment not only ameliorated histopathological damage and pulmonary edema but also improved pulmonary mechanics and reduced inflammatory cell infiltration, indicating a coordinated preservation of alveolar integrity and lung function. In addition, our findings suggest that the microRNA cargo of GiEVs, particularly miR-339-3p, may contribute to these therapeutic effects. Notably, these results extend our previous work demonstrating the efficacy of human placental MSC–derived EVs in the same model [9], and further support the advancement of EV-based therapies toward more scalable and translationally feasible platforms.

Mechanistically, our data suggest that the therapeutic effects of GiEVs are mediated through coordinated modulation of inflammatory, oxidative, and apoptotic pathways. GiEV administration suppressed NF-κB signaling and macrophage activation, leading to reduced downstream pro-inflammatory cytokine production and decreased leukocyte infiltration. In parallel, GiEVs attenuated oxidative stress, as evidenced by reduced lipid peroxidation and dampening of the stress-induced antioxidant response, reflected by decreased MDA levels and HO-1 expression. Notably, HO-1 expression was elevated in the ALI group—likely representing a compensatory response to oxidative stress—GiEV treatment reduced both MDA and HO-1 levels, indicating an overall reduction in oxidative stress burden rather than simple activation of antioxidant pathways. In addition, GiEVs limited apoptosis, as demonstrated by reduced TUNEL positivity and decreased expression of pro-apoptotic cleaved caspase-3. Collectively, these findings support a model in which GiEVs interrupt the self-amplifying cycle of inflammation, oxidative injury, and cell death that drives the progression of aspiration-induced lung injury. The proposed mechanisms and overall therapeutic effects of GiEVs are summarized in Figure 8.

A key finding of this study is the identification of miR-339-3p as a critical mediator of GiEV function. Loss-of-function experiments using inhibitor-treated GiEVs demonstrated that suppression of miR-339-3p largely abrogated the protective effects on lung injury severity and pulmonary edema, indicating a miR-339-3p–dependent mechanism. This provides direct functional evidence linking EV cargo to biological activity in vivo. Although the precise downstream targets remain to be fully elucidated, our findings, together with prior evidence, suggest that miR-339-3p may primarily regulate inflammatory and apoptotic pathways. Notably, previous work has identified TNF receptor–associated factor 3 (TRAF3) as a direct target of miR-339-3p, where miR-339-3p overexpression suppressed pro-inflammatory cytokine production (TNF-α, IL-1β, and IL-6) and reduced apoptosis in a caerulein-induced acute pancreatitis model [26]. Given the central role of TRAF3 in regulating NF-κB signaling pathways [27], this mechanism provides a plausible explanation for the attenuation of inflammatory and apoptotic responses observed in our study. In addition, emerging clinical evidence suggests that reduced miR-339-3p expression is associated with oxidative imbalance [28], although its causal role in regulating oxidative stress remains to be clarified. Taken together, these findings support a model in which miR-339-3p contributes to the anti-inflammatory and anti-apoptotic effects of GiEVs, while its involvement in redox regulation warrants further investigation.

Importantly, the present study builds directly upon our recent work demonstrating that human placental MSC–derived EVs effectively mitigate aspiration-induced ALI in the same experimental model [9]. While that study established proof-of-concept for EV-based therapy in aspiration-induced lung injury, it also highlighted key translational challenges inherent to mammalian cell–derived EVs, including limited scalability, donor-dependent variability, and manufacturing complexity. In this context, the current study represents a logical next step, shifting from proof-of-efficacy to evaluation of a more translationally feasible EV platform. Plant-derived EVs, such as GiEVs, offer several advantages that directly address these limitations. They can be produced in large quantities from readily available plant sources, enabling scalable and cost-effective manufacturing without the need for cell expansion [12, 13]. Their non-mammalian origin reduces concerns related to pathogen transmission, donor variability, and ethical constraints, while their physicochemical stability supports storage and distribution [12, 13]. Notably, despite their distinct biological origin, GiEVs retained robust therapeutic efficacy in vivo, supporting the concept that plant-derived EVs can functionally recapitulate key immunomodulatory effects of mammalian EVs.

Despite these advantages, differences in biological potency should be considered. In this study, a higher dose of GiEVs—approximately one order of magnitude greater than that required for human placental MSC-derived EVs in the same model [9] (1×10^9 vs. 1×10^8 particles per mouse)—was needed to achieve significant therapeutic effects. This difference may reflect variations in cargo composition, target specificity, and cellular uptake efficiency between plant- and mammalian-derived vesicles. Notably, human placental MSC-derived EVs are enriched in microRNAs such as let-7i-5p, which directly targets toll-like receptor 4 (TLR4) to suppress inflammatory signaling [29], thereby exerting broad immunomodulatory effects. In contrast, GiEVs are enriched in miR-339-3p, which may regulate more specific signaling nodes, such as TRAF3, involved in inflammatory and apoptotic pathways. These differences in microRNA cargo may influence the breadth and potency of downstream biological effects. In addition, cross-kingdom differences in microRNA processing and target engagement may further impact the efficiency of plant-derived vesicles in mammalian systems.

Importantly, this increased dosing requirement did not compromise safety, as no significant alterations in liver or renal function markers, including alanine aminotransferase, aspartate aminotransferase, blood urea nitrogen, and creatinine, were observed even at this or higher doses (namely, 1×10^9 and 1×10^{10} particles per mouse) (Supplementary Figure 1B). Taken together, these findings suggest a translational paradigm in which plant-derived EVs trade modestly reduced per-particle potency for substantial gains in scalability, safety margin, and manufacturing feasibility. This balance may be particularly advantageous in acute inflammatory conditions, where rapid availability and the ability to administer higher doses safely are critical.

Several limitations warrant consideration. First, although miR-339-3p was identified as a key functional mediator, its direct molecular targets and signaling networks remain to be fully defined. Second, while intratracheal administration enabled controlled delivery in this model, clinically relevant routes such as aerosolized or systemic delivery should be explored. Third, the specific cellular targets of GiEVs within the lung were not characterized. Fourth, this study was conducted exclusively in male mice; therefore, potential sex-specific differences in response to GiEV treatment were not evaluated. Given that sex hormones and immune responses are known to influence inflammatory and oxidative pathways in acute lung injury [30], future studies incorporating both sexes will be important to determine the generalizability of these findings. Finally, this study focused on the acute phase of lung injury, and the long-term effects of GiEV treatment, including roles in tissue repair and fibrosis, remain to be determined. In addition, standardization of plant-derived EV isolation and characterization will be essential for reproducibility and clinical translation.

Together with our prior work on human placental MSC-derived EVs, these findings support a translational progression from experimental efficacy to clinically scalable therapy, highlighting plant-derived EVs as a practical and deployable strategy for managing aspiration-induced lung injury in perioperative and critical care settings.

In conclusion, GiEVs mitigate aspiration-induced lung injury through coordinated suppression of inflammation, oxidative stress, and apoptosis, supporting their therapeutic potential in acute lung injury. Notably, miR-339-3p contributes, at least in part, to these protective effects, as evidenced by its role in modulating lung injury severity and pulmonary edema.

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⑧ **Table:** Not applicable

⑨ Legends for figures and photographs

Figure 1. Characterization and biodistribution of ginger-derived extracellular vesicles (GiEVs). (A) Transmission electron microscopy (TEM) image showing the characteristic round or cup-shaped morphology of GiEVs. Scale bar = 200 nm. (B) Nanoparticle tracking analysis (NTA) showing the size distribution and concentration profile of isolated GiEVs. (C) Zeta potential of isolated GiEVs, indicating their surface charge. (D) In vivo fluorescence imaging using an in vivo imaging system (IVIS) demonstrating the biodistribution of Cy7-labeled GiEVs following intratracheal administration at 0, 2, 24, and 48 hours (n = 3 per time point). Quantitative analysis of fluorescence signal intensity in the lung region over time is shown. Data are presented as mean ± standard deviation. *P < 0.05 vs. 0 hours.

Figure 2. Ginger-derived extracellular vesicles (GiEVs) attenuate lung injury and improve pulmonary function in aspiration-induced acute lung injury (ALI). (A) Representative hematoxylin and eosin (H&E) staining of lung tissues and quantitative lung injury scores across experimental groups. Data are from 6 mice per group. (B) Pulmonary function parameters, including inspiratory volume, expiratory volume, forced expiratory volume in 0.1 second (FEV0.1), forced expiratory volume, dynamic compliance, airway resistance, and elastance across experimental groups. Data are from 6 mice per group. Sham: the Sham-operation group, GiEV: the GiEVs only group. ALI: the aspiration-induced ALI group. ALI+GiEV: the ALI treated with GiEVs group. Data are presented as mean ± standard deviation. *P < 0.05, ALI group vs. Sham group; #P < 0.05 ALI+GiEV group vs. ALI group.

Figure 3. Ginger-derived extracellular vesicles (GiEVs) suppress pulmonary inflammation. (A) Lung wet-to-dry weight ratio as an indicator of pulmonary edema. Data are from six mice per group. (B) Total and differential leukocyte counts in bronchoalveolar lavage fluid (BALF), including white

blood cells (WBC), neutrophils, lymphocytes, and monocytes. Data are from six mice per group. Sham, the Sham-operation group; GiEV, the GiEV-only group; ALI, the aspiration-induced acute lung injury group; ALI+GiEV, the ALI treated with GiEV group. Data are presented as mean ± standard deviation. *P < 0.05, ALI group vs. Sham group; #P < 0.05, ALI+GiEV group vs. ALI group.

Figure 4. Ginger-derived extracellular vesicles (GiEVs) inhibit inflammatory signaling and macrophage activation. (A) Immunohistochemical staining and quantification of phosphorylated nuclear factor-κB (p-NF-κB) and inducible nitric oxide synthase (iNOS) in lung tissues. Brown staining indicates positive expression, with representative positive cells highlighted by red arrows. Data are from six mice per group. (B) Immunoblotting analysis and quantification of p-NF-κB and iNOS expression, with Actin used as the internal control. Data are from four mice per group. (C) Levels of pro-inflammatory cytokines, including tumor necrosis factor-α (TNF-α), interleukin-1β (IL-1β), and interleukin-6 (IL-6), in lung tissues. Data are from six mice per group. Sham, the Sham-operation group; GiEV, the GiEV-only group; ALI, the aspiration-induced acute lung injury group; ALI+GiEV, the ALI treated with GiEVs group. Data are presented as mean ± standard deviation. *P < 0.05, ALI group vs. Sham group; #P < 0.05, ALI+GiEV group vs. ALI group.

Figure 5. Ginger-derived extracellular vesicles (GiEVs) attenuate pulmonary oxidative stress. (A) Immunohistochemical staining and quantification of malondialdehyde (MDA) and heme oxygenase-1 (HO-1) in lung tissues. Brown staining indicates positive expression, with representative positive cells highlighted by red arrows. Data are from six mice per group. (B) Immunoblotting analysis and quantification of MDA and HO-1 protein expression, with Actin used as the internal control. Data are from four mice per group. Sham, the Sham-operation group; GiEV, the GiEV-only group; ALI, the aspiration-induced acute lung injury group; ALI+GiEV, the ALI treated with GiEVs group. Data are

presented as mean ± standard deviation. *$P < 0.05$, ALI group vs. Sham group; #$P < 0.05$, ALI+GiEV group vs. ALI group.

Figure 6. Ginger-derived extracellular vesicles (GiEVs) reduce apoptosis in aspiration-induced acute lung injury. (A) Terminal deoxynucleotidyl transferase dUTP nick end labeling (TUNEL) staining and quantification of apoptotic cells in lung tissues. Bright green fluorescence indicates TUNEL-positive staining, with representative positive cells highlighted by red arrows. Data are from six mice per group. (B) Immunoblotting analysis and quantification of cleaved caspase-3 expression, with Actin used as the internal control. Data are from four mice per group. Sham, the Sham-operation group; GiEV, the GiEV-only group; ALI, the aspiration-induced acute lung injury group; ALI+GiEV, the ALI treated with GiEVs group. Data are presented as mean ± standard deviation. *$P < 0.05$, ALI group vs. Sham group; #$P < 0.05$, ALI+GiEV group vs. ALI group.

Figure 7. MicroRNA-339-3p (miR-339-3p) contributes to the therapeutic effects of ginger-derived extracellular vesicles (GiEVs). (A) Heatmap showing the microRNA expression profile in GiEV preparations. (B) Representative hematoxylin and eosin (H&E) staining and quantitative lung injury scores across experimental groups. Data are from six mice per group. (C) Lung wet-to-dry weight ratio indicating pulmonary edema across treatment groups. Data are from six mice per group. ALI, the aspiration-induced acute lung injury group; ALI+GiEV, the ALI treated with GiEVs group. ALI+GiEVi: the ALI treated with miR-339-3p inhibitor-loaded GiEVs. Data are presented as mean ± standard deviation. *$P < 0.05$, ALI+GiEV group vs. ALI group; †$P < 0.05$, ALI+GiEVi group vs. ALI+ALI+GiEV group.

Figure 8. Schematic illustration depicting the pathophysiological mechanisms of aspiration-induced acute lung injury and the therapeutic actions of ginger-derived extracellular vesicles (GiEVs) in mice. NF-κB: nuclear factor-κB; iNOS: inducible nitric oxide synthase; TNF-α: tumor necrosis

factor- α; IL-1β: interleukin-1β; IL-6: interleukin-6; HO-1: heme oxygenase-1; MDA: malondialdehyde; miR-339-3p: microRNA-339-3p.

Figure

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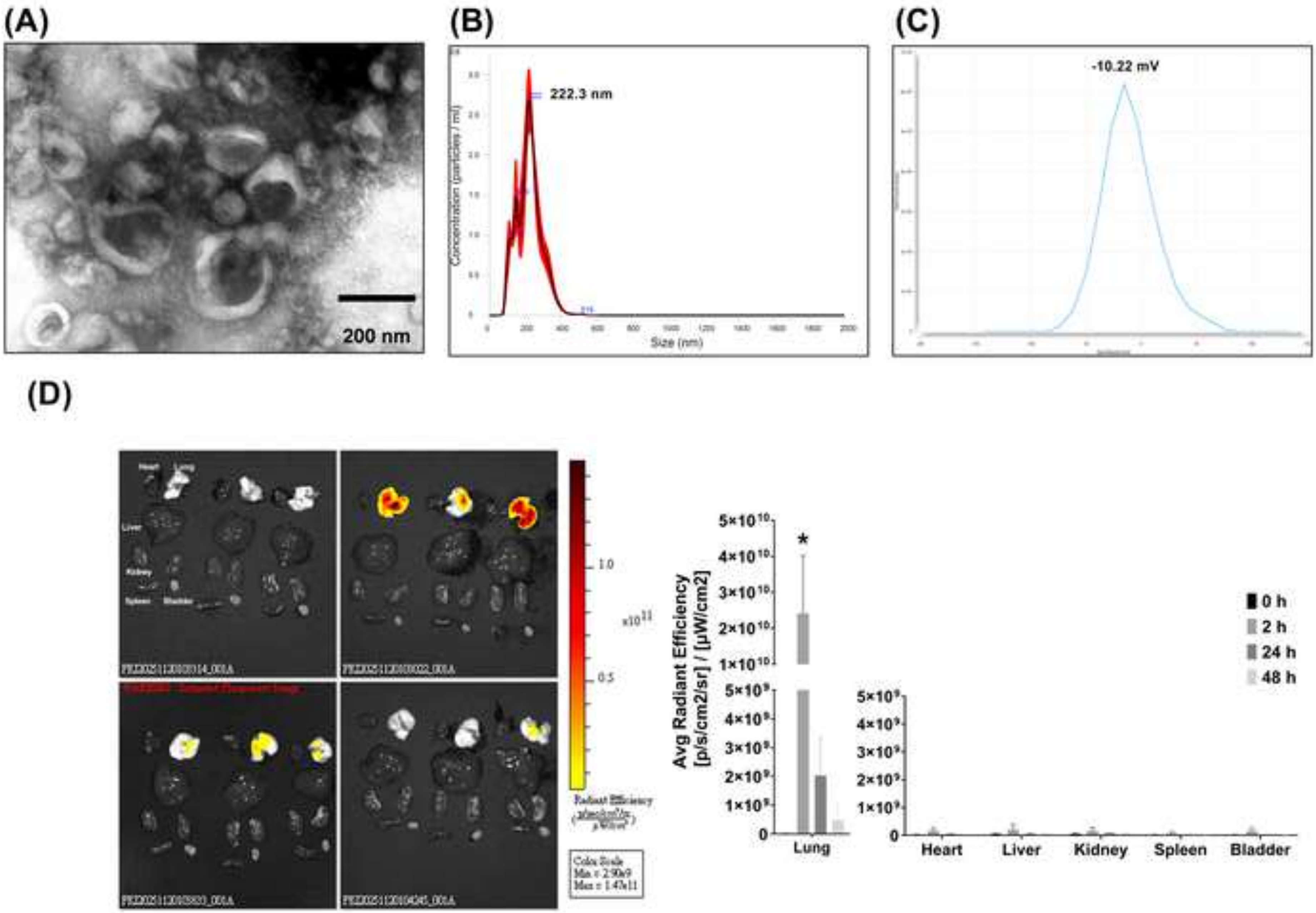
Figure 1



Figure

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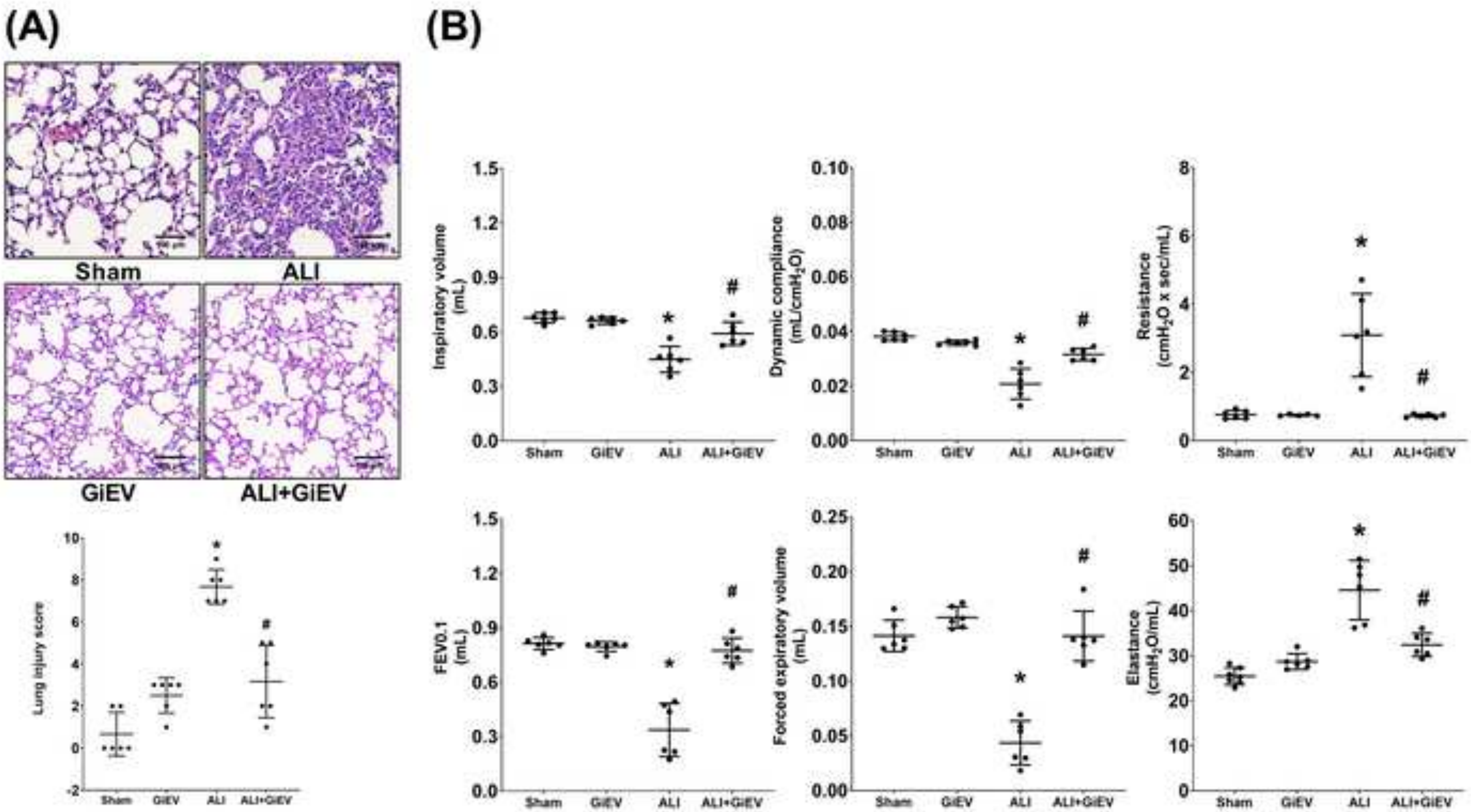
Figure 2



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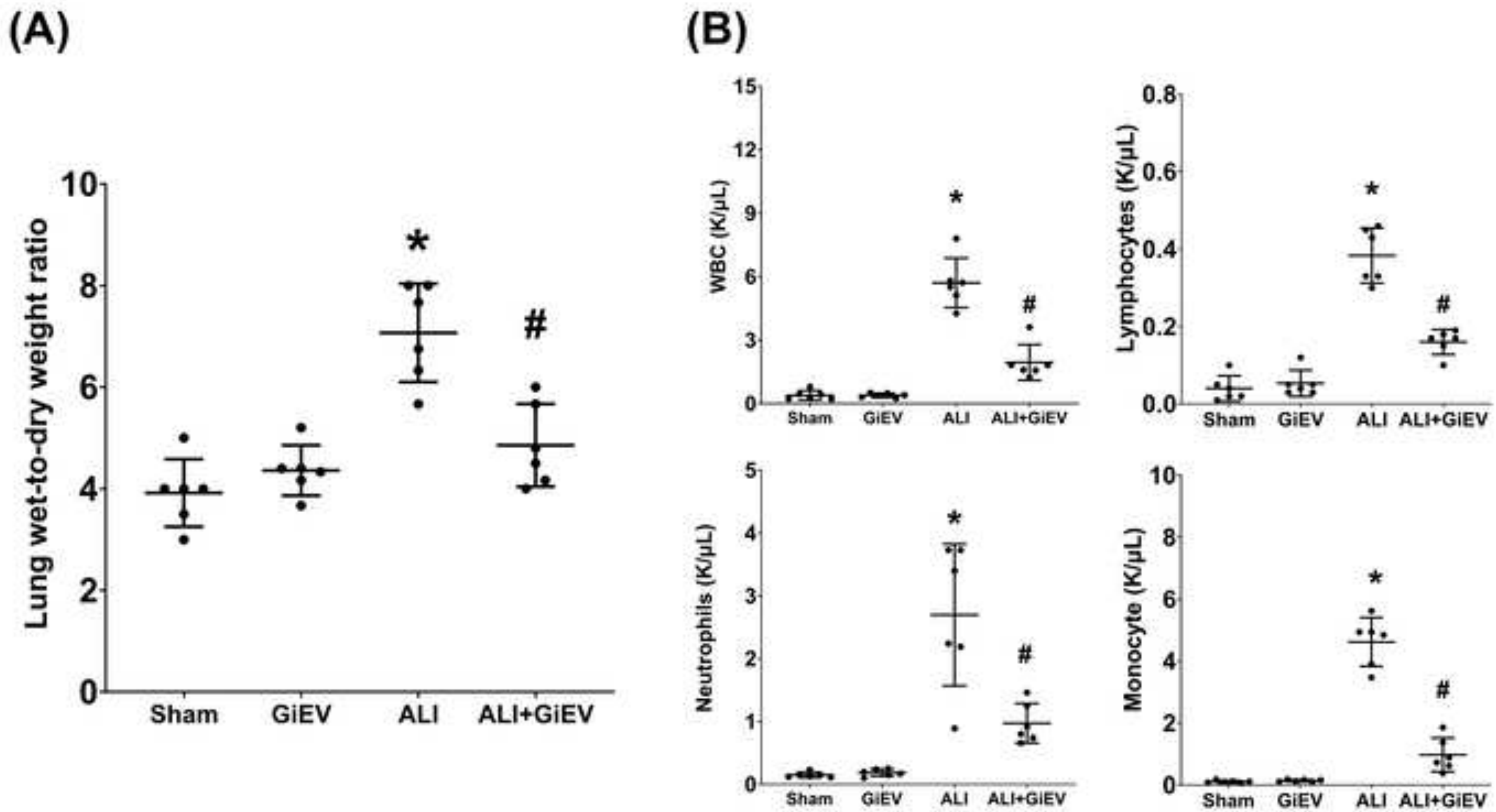

(A)
Lung wet-to-dry weight ratio
10
8
6
4
2
0
*
#
Sham
GiEV
ALI
ALI+GiEV
(B)
WBC (K/μL)
15
12
9
6
3
0
Lymphocytes (K/μL)
0.8
0.6
0.4
0.2
0.0
Neutrophils (K/μL)
5
4
3
2
1
0
Monocyte (K/μL)
10
8
6
4
2
0
*
#
Sham
GiEV
ALI
ALI+GiEV

Figure 4

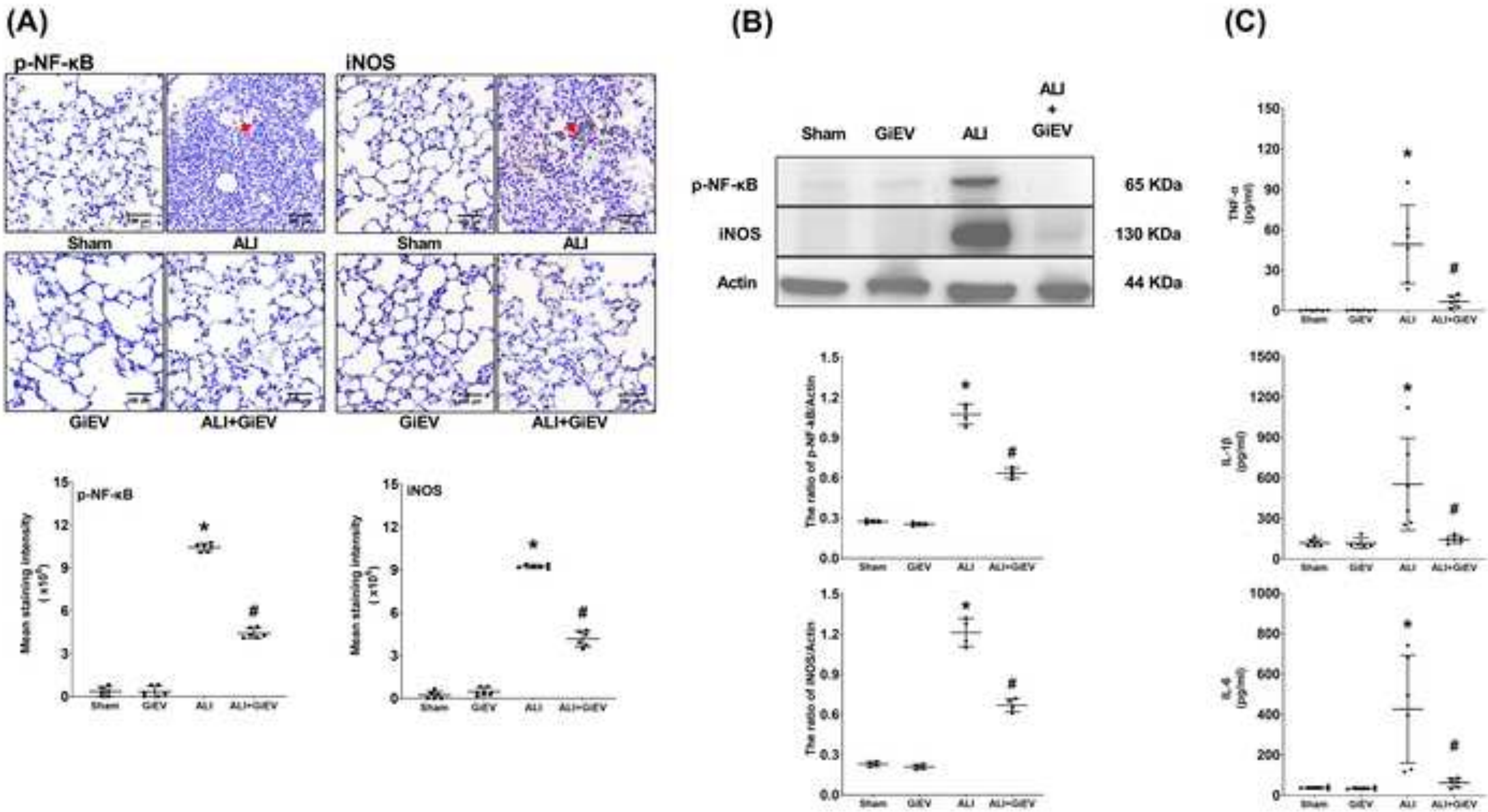
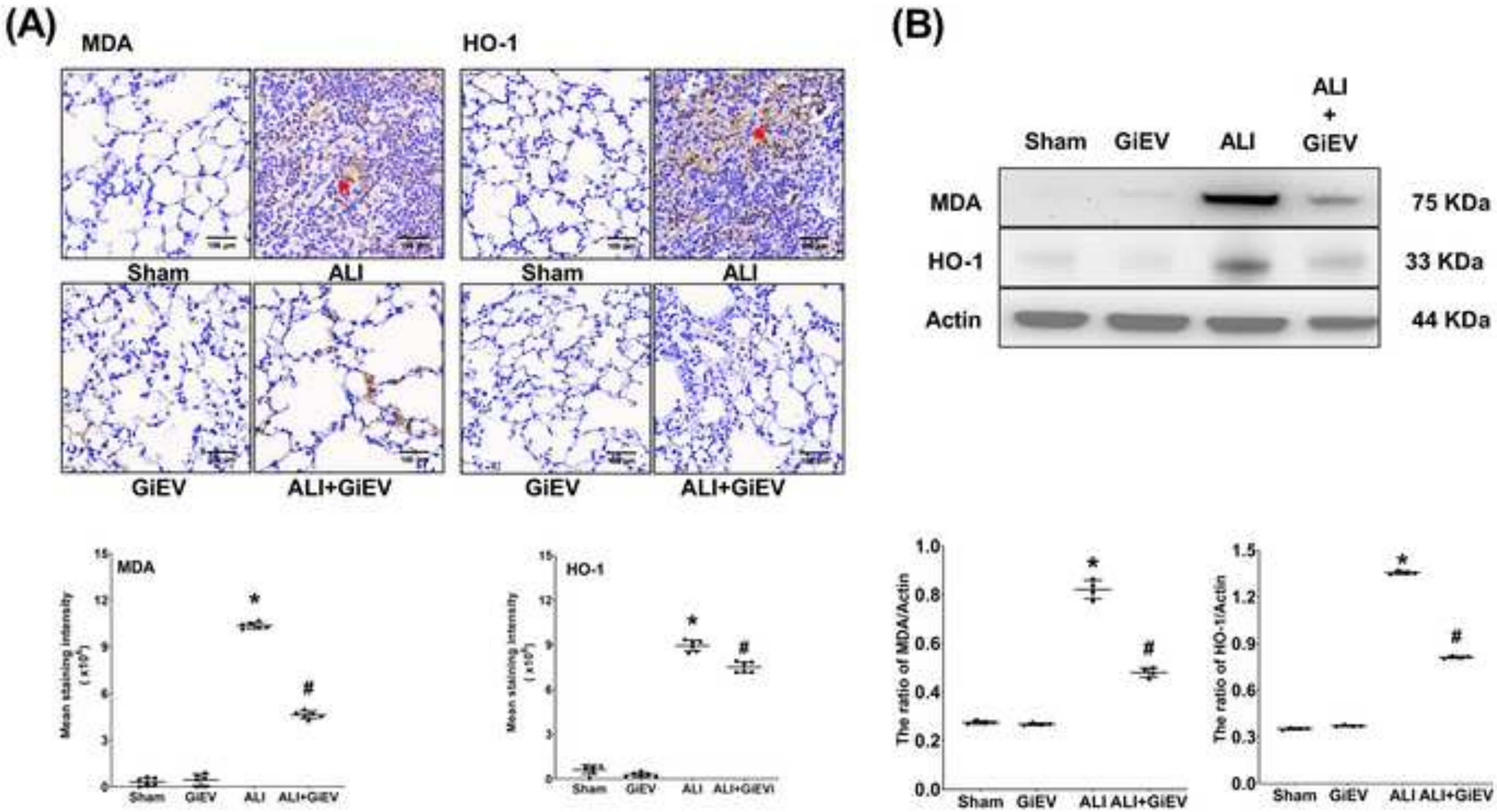

(A)
p-NF-κB
iNOS
Sham
ALI
GiEV
ALI+GiEV
Mean staining intensity (x10^5)
(B)
Sham
GiEV
ALI
ALI + GiEV
p-NF-κB
iNOS
Actin
65 KDa
130 KDa
44 KDa
The ratio of p-NF-κB/Actin
The ratio of iNOS/Actin
(C)
TNF-α (pg/ml)
IL-1β (pg/ml)
IL-6 (pg/ml)

Figure 5


(A)
MDA
HO-1
Sham
ALI
GiEV
ALI+GiEV
Mean staining intensity
(x10³)
(B)
Sham
GiEV
ALI
ALI
+
GiEV
MDA
HO-1
Actin
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33 KDa
44 KDa
The ratio of MDA/Actin
The ratio of HO-1/Actin
ALI+GiEV

Figure 6

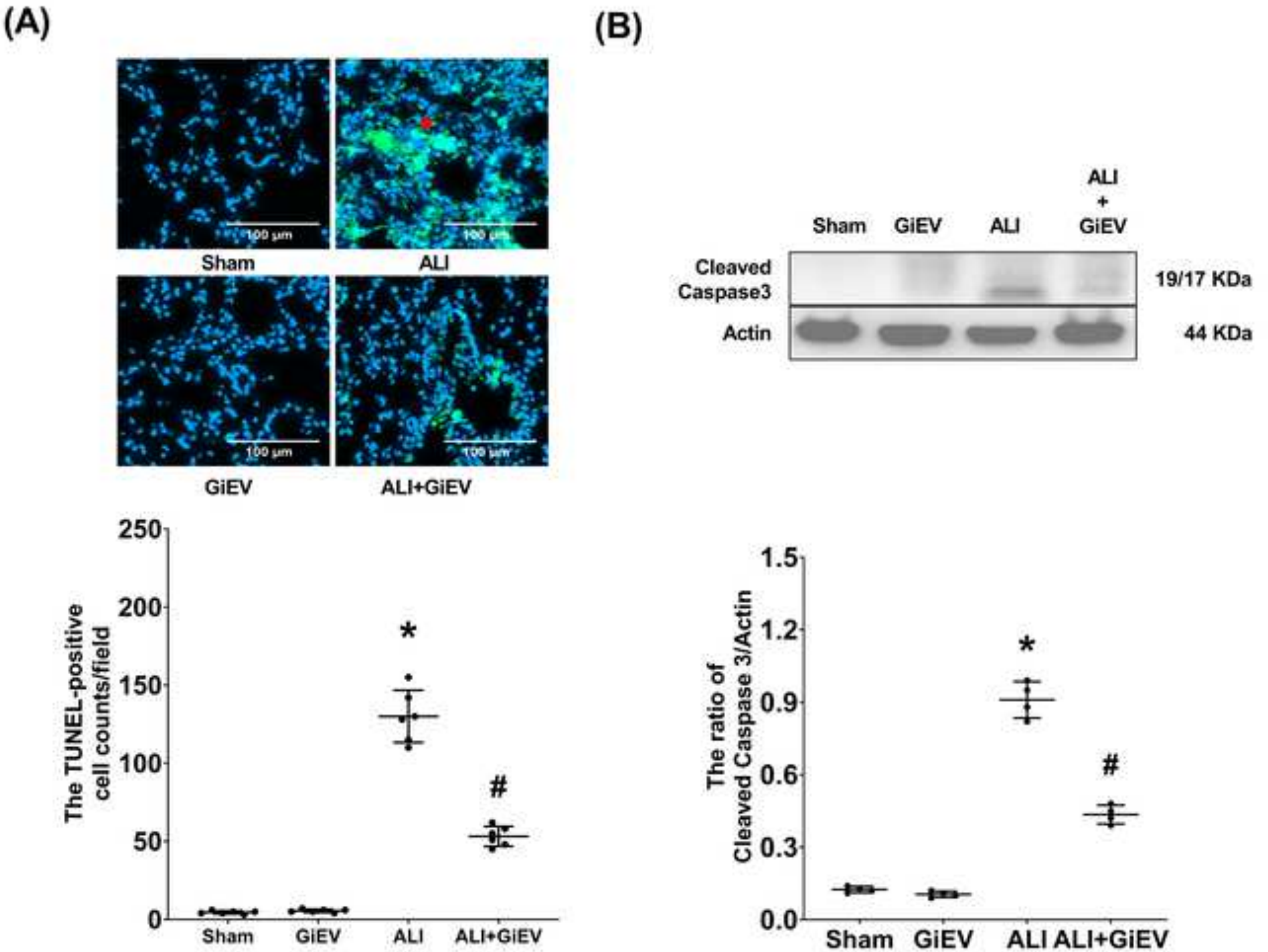
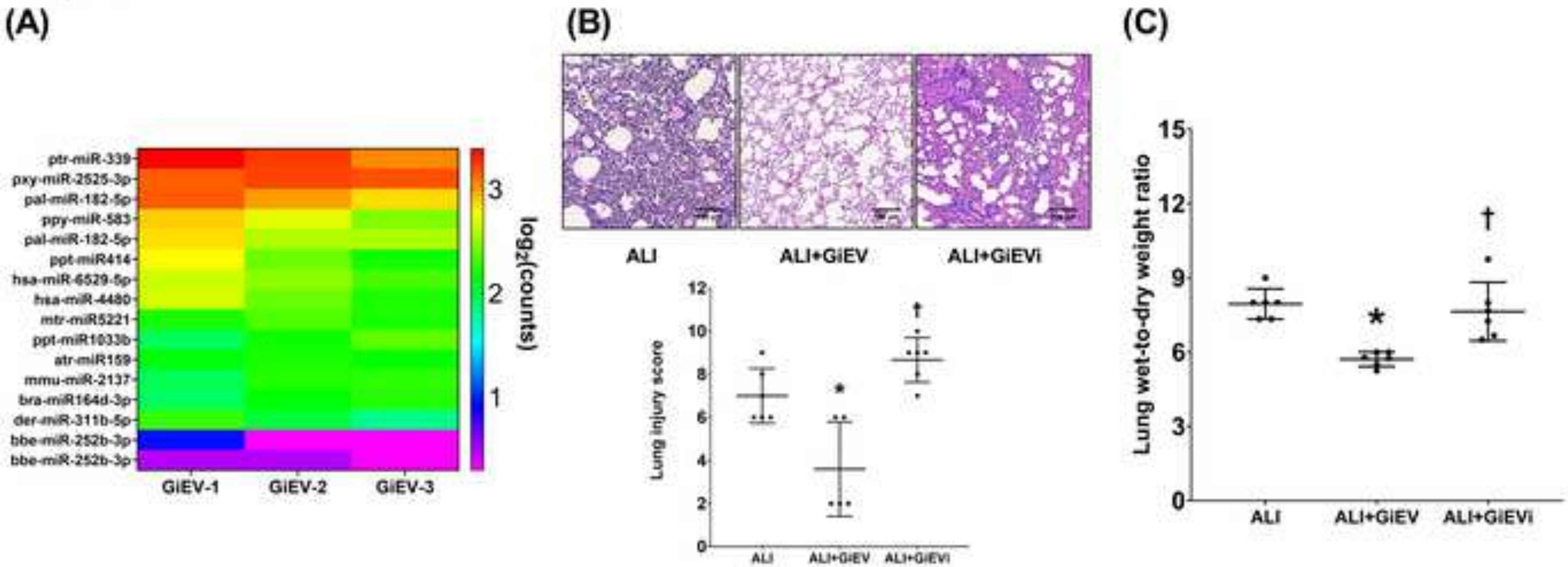

(A)
100 μm
Sham
ALI
GiEV
ALI+GiEV
The TUNEL-positive cell counts/field
250
200
150
100
50
0
*
#
Sham
GiEV
ALI
ALI+GiEV
(B)
Sham
GiEV
ALI
ALI + GiEV
Cleaved Caspase3
19/17 KDa
Actin
44 KDa
The ratio of Cleaved Caspase 3/Actin
1.5
1.2
0.9
0.6
0.3
0.0
*
#
Sham
GiEV
ALI
ALI+GiEV

Figure 7


(A)
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pal-miR-182-5p
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pal-miR-182-5p
ppt-miR414
hsa-miR-6529-5p
hsa-miR-4480
mtr-miR5221
ppt-miR1033b
atr-miR159
mmu-miR-2137
bra-miR164d-3p
der-miR-311b-5p
bbe-miR-252b-3p
bbe-miR-252b-3p
GiEV-1
GiEV-2
GiEV-3
3
2
1
log2(counts)
(B)
ALI
ALI+GiEV
ALI+GiEVi
Lung injury score
0
2
4
6
8
10
12
*
†
ALI
ALI+GiEV
ALI+GiEVi
(C)
Lung wet-to-dry weight ratio
0
3
6
9
12
15
*
†
ALI
ALI+GiEV
ALI+GiEVi

Figure

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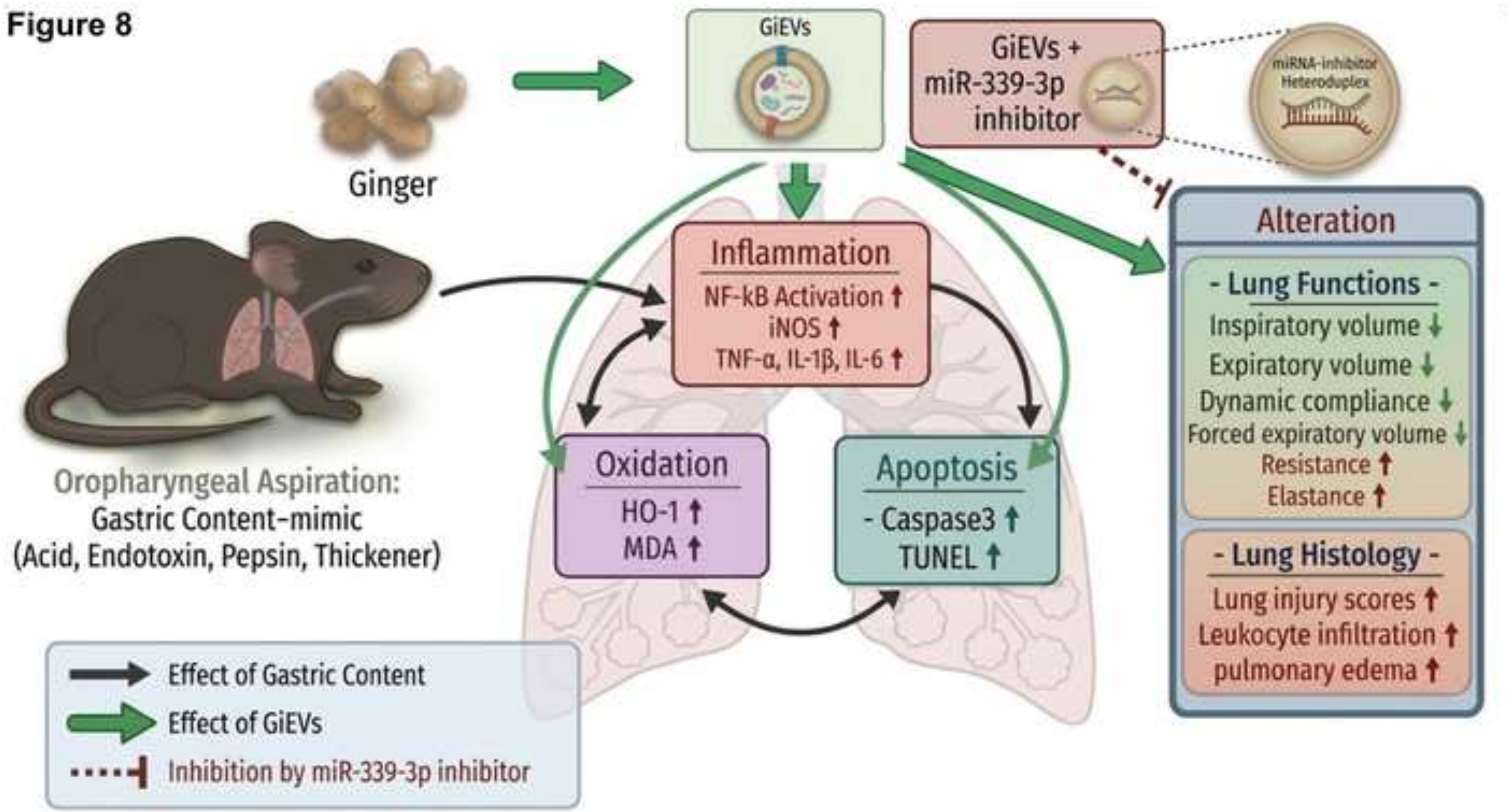

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